Chapter 5

Dictionary & Sets

Dictionary is a collection of key-value pairs

Syntax:

```
a={ "Key" : "Value"
    "Jack" : "Code"
    "marks": "100"
    "list": "[1,2,9]"}
```

(Key → "Key", Values → "Value")

a["key"] ⇒ Print "Values"

a["list"] ⇒ Prints [1,2,9]

Properties of a Python Dictionaries

1> It is unordered
2> It is mutable
3> It is indexed
4> Cannot contain duplicate keys

Dictionary Methods - Consider the following diction

```
a = {"name" : "Harry"
     "from" : "India"
     "marks": "[92,98,96]"}
```

1> a.items(): returns a list of (key, value) tuples.
2> a.keys(): returns a list containing dictionary keys.
3> a.update({"friend":"Sam"}): Updates the dictionary with supplied key-value pairs.
4> a.get("name"): (Mor) returns the value of the specified keys.

More methods are available on docs.python.org.

Sets in Python - Set is a collection of non repetitive elements

```
S = Set()
S.add(1)
S.add(2)
```

⇒ No repetion allowed!

If you are a programming beginner without muc knowledge of mathematical operations on sets, you can simply look at sets in Python as data types continuing unique values.

Properties of Sets

1> Sets are unordered. ⇒ Element's order doesn't matte
2> Sets are unindexed. ⇒ Cannot access element by inde
3> There is no way to change items in sets.
4> Sets cannot contain duplicate values.

operations on Sets - Consider the followind set

$S = \{1, 8, 2, 3\}$

1> len(S): Returns 4, the length of set.
2> S.remove(8): removes 8 from S
3> S.pop(): Removes an arbitrary element from the set and return value
4> S.clear(): Empty the sets.
5> S.union({8, 11}): Returns a new set with all items from both sets ⇒ {1, 8, 2, 3, 11}
6> S.intersection({8, 11}): Returns a set which contains only items in both sets. ⇒ {8}

Chapter 5 : Practice Set

1. Write a program to create a dictionary of Hindi words with values as their english translation. Provide user with an option to look it up!

2. Write a program to input eight numbers from the user and display all the unique numbers.

3. Can we have a set with 18 (int) and "18" (str) as a values in it?

4. What will be the length of following set S:

```
S = Set()
S.add(20)
S.add(20.0)
S.add("20")
```

⇒ Length of S after these operations

5. S = { } - What is the type of S?

6. Create an empty dictionary. Allow 4 friends to enter their favourite language as values and use keys as their names. Assume that the names are unique.

7. If names of 2 friends are same; what will to the program in Problem 6?

8. If language of two friends are same. What will happen to the program in problem 6?

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EXPLORER

CHAPTER-5_DICT_&_SETS

- dict.py
- methods.py
- practice_set5.ipynb
- Set_0.py
- set_methods.py
- set_union_intersection.py

OUTLINE

TIMELINE

dict.py > marks

```python
marks={"virendra": 98,
    "ajay": 87,
    "sachin": 78,
    "list":[1,2,3,4]}
print(marks,type(marks))
# Accessing values using keys
print(marks["virendra"])
print(marks["ajay"])
print(marks["list"][2])
```

methods.py > marks

```python
marks={"virendra": 98,
    "ajay": 87,
    "sachin": 78,
    "list":[1,2,3,4]}
# dictionary methods
#print(marks.values())
marks.update({"virendra":90,"jaya":85}) # updating
#print(marks.keys())
print(marks.get("vire"))  # accessing value using
print(marks["virendra"]) # accessing value using k
print(marks.pop("ajay")) # removes key-value pair
print(marks.popitem()) # removes last inserted key
```

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EXPLORER

CHAPTER-5_DICT_&_SETS

- 1.png
- dict.py
- methods.py
- practice_set5.ipynb
- Set_0.py
- set_methods.py
- set_union_intersection.py

OUTLINE

TIMELINE

set_methods.py > ...

```python
set1=set() #Creating an empty set
set1.add(3) #Adding element to set
print(set1)
set1.add("virendra")
print(set1)
set1.update([1,2,3,4]) #Adding multiple elem
print(set1)
set1.remove(3) #Removing element from set
print(set1)
set1.discard("virendra") # No Error Removing
print(set1)
print(len(set1)) #Finding length of set
print(set1.pop()) #Removing and returning an
```

set_union_intersection.py > s1

```python
s1={1,2,3,4,5}
s2={4,5,6,7,8}
print(s1.union(s2))
print(s1.intersection(s2))
print(s1.issubset(s2))
print(s1.difference(s2))
```

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